

PUNTA CANA INTL, Dominican Republic

WMO: 784790

Lat: 18.567N

Lon: 68.363W

Elev: 14

StdP: 101.15

Time Zone: -4.00 (W04)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.8	20.8	16.2	11.6	23.7	17.2	12.3	23.8	9.1	25.2	8.4	25.3	0.4	0	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	32.1	26.6	31.6	26.5	31.1	26.4	27.4	30.6	27.1	30.4	26.7	30.1	5.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	29.5	26.1	21.6	29.4	25.9	21.3	29.3	86.8	31.3	85.6	30.6	84.1	30.3	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	8.0	7.4	16.7	32.9	1.5	1.1	15.6	33.7	14.8	34.4	13.9	35.0	12.8	35.8		
(5)			16.2	28.1	1.5	0.5	15.1	28.5	14.3	28.8	13.4	29.0	12.4	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	25.1	25.2	25.3	26.2	26.8	27.6	28.3	28.3	27.9	27.4	26.4	25.8		
	DBStd	1.52	1.10	1.02	1.11	0.96	0.95	0.85	0.90	0.97	0.93	0.99	1.03	1.16		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6092	467	426	474	486	520	529	566	566	537	539	493	489		
	CDD18.3	3051	208	193	216	236	262	279	308	308	287	281	243	231		
	CDH23.3	29758	1498	1368	1599	2055	2560	3060	3606	3642	3256	2956	2243	1915		
	CDH26.7	8379	191	188	272	432	651	970	1289	1362	1189	976	542	317		
(14) Wind	WSAvg	3.8	4.0	4.1	4.1	4.1	3.6	3.5	4.2	4.1	3.2	2.9	3.4	4.3		
(15) Precipitation	PrecAvg	1072	65	54	56	70	123	107	80	93	99	146	103	78		
	PrecMax	1661	152	175	158	222	349	208	206	271	252	564	277	183		
	PrecMin	659	12	4	9	1	7	9	23	24	8	36	21	14		
	PrecStd	250	33	39	39	56	87	54	39	58	57	107	66	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.9	30.2	30.8	31.2	32.0	32.2	32.2	32.2	32.1	30.9	30.1		
		MCWB	24.0	24.3	24.7	25.9	26.5	26.8	26.7	26.8	26.7	26.4	26.3	24.8		
	2%	DB	28.6	28.9	29.2	30.0	30.7	31.1	31.8	31.9	32.0	31.2	30.1	29.2		
		MCWB	23.8	23.6	23.9	25.1	26.1	26.4	26.6	26.7	26.5	26.3	25.7	24.5		
	5%	DB	28.1	28.2	28.8	29.2	30.0	30.8	31.1	31.2	31.2	31.0	29.9	28.9		
		MCWB	23.6	23.3	23.6	24.3	25.5	26.3	26.3	26.5	26.5	26.3	25.5	24.4		
	10%	DB	27.9	27.9	28.1	28.9	29.2	30.1	30.9	31.0	31.0	30.2	29.2	28.2		
		MCWB	23.5	23.2	23.0	24.1	25.1	25.9	26.2	26.4	26.5	25.9	25.0	24.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.3	25.3	26.7	27.6	27.8	27.6	27.7	27.7	27.3	26.7	26.0		
		MCDB	27.6	28.0	29.1	30.0	30.5	30.6	31.0	31.0	30.7	30.4	29.5	28.4		
	2%	WB	24.7	24.6	24.6	25.8	26.6	27.0	27.0	27.2	27.2	26.9	26.4	25.4		
		MCDB	27.5	27.6	28.1	28.8	29.6	30.2	30.5	30.6	30.5	30.1	29.2	28.1		
	5%	WB	24.2	24.0	24.1	25.3	26.1	26.6	26.7	26.8	26.8	26.6	26.0	24.9		
		MCDB	27.2	27.3	27.6	28.2	29.0	29.7	30.1	30.3	30.2	29.8	28.8	27.9		
	10%	WB	23.8	23.6	23.7	24.7	25.6	26.3	26.4	26.6	26.6	26.3	25.6	24.5		
		MCDB	26.8	26.8	27.1	27.6	28.5	29.3	29.7	30.0	29.9	29.5	28.4	27.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	5.8	6.0	5.8	5.8	5.7	5.5	5.8	6.5	6.6	5.9	5.5		
		MCDBR	6.2	6.1	6.7	6.1	6.2	6.1	5.8	6.0	6.9	7.1	6.3	5.9		
		MCWBR	3.4	3.2	3.2	2.9	3.1	3.0	2.7	2.9	3.4	3.4	3.4	3.1		
	5% WB	MCDBR	6.0	6.0	6.3	6.0	6.0	5.9	5.5	5.9	6.6	6.6	5.9	5.5		
		MCWBR	3.6	3.4	3.3	3.3	3.3	3.3	2.8	3.0	3.4	3.5	3.4	3.2		
(40) Clear-Sky Solar Irradiance	taub		0.366	0.380	0.383	0.410	0.441	0.513	0.507	0.482	0.428	0.387	0.388	0.381		
	taud		2.568	2.521	2.506	2.461	2.394	2.139	2.158	2.255	2.445	2.567	2.559	2.548		
	Ebn at Noon		918	921	926	897	859	791	797	824	871	903	889	891		
	Edh at Noon		95	104	109	115	122	156	153	140	115	98	95	95		
(44) All-Sky Solar Radiation	RadAvg		4.60	5.36	5.95	6.24	6.04	6.14	6.33	6.11	5.70	5.09	4.51	4.32		
	RadStd		0.18	0.14	0.22	0.35	0.38	0.30	0.27	0.24	0.38	0.37	0.34	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.27	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page